

Technology #05: Phage Anti-Biofilm Coating for Shunt Infection Prevention

Technology Domain: Microbiology + Surface Science + Biomaterials

Mechanism: CSF shunt catheter with *S. aureus* phage K immobilized on electrospun titanium-coated surface; phages prevent biofilm formation by infecting and lysing *S. aureus* before biofilm establishes

Maturity: ENGINEERING_DEFINITION

Current State: No physical prototype. Computational model and engineering definition exist.

Key Risk: Phage K cannot retain infectivity when immobilized on Ti surface for >24h

Next Experiment: Ti-coated catheter coupons (various electrospinning parameters)

Buyer Receives: Phage K + Ti coating concept + mechanism description; Phage adsorption + Monod biofilm governing equations; Bench test protocol (ASTM E2180, ISO 22196, titer assay); External precedent catalog (phage K, Ti coatings); Critical UNKNOWN disclosure (phage stability, immune response, resistance); This engineering dossier with failure analysis + build plan

Buyer Must Build: cGMP-grade phage K manufacturing process; Electrospun Ti coating process for catheter geometry; Phage immobilization chemistry + QC specs; Aseptic assembly process (terminal sterilization likely incompatible); Regulatory submission (likely combination product — biologics + device; complex pathway); Clinical validation (infection rate endpoint); Long-term stability + resistance evolution monitoring

Transfer Posture: Sponsored Validation

Target Buyer: Anti-infection medical device company; neuroshunt manufacturer

CONFIDENTIAL. See full engineering dossier for details.